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Paper: Enterprise Systems

Course Code: SI-342

Q1. Answer the following short questions: (6x5=30)

Q1 What is Data Set in ADO.NET?

Answer:

1. A **DataSet** is a collection of data stored in memory.
2. It works like a **mini database** inside your program.
3. It can store multiple tables at the same time.
4. It is **disconnected**, meaning it does not need a constant database connection.
5. It is used to **read, store, and manipulate data** in applications.

Q2 What are SQL connection methods?

Answer:

1. **Open()** → Used to open the database connection.
2. **Close()** → Used to close the connection.
3. **ExecuteReader()** → Reads data from database.
4. **ExecuteNonQuery()** → Used for insert, update, delete.
5. **ExecuteScalar()** → Returns a single value (like count).
- 6.

Q3 How to detect a mobile device in jQuery?

Answer:

1. Use **navigator.userAgent** to check device type.
2. Check if it contains words like **Android, iPhone**.
3. Use jQuery condition to apply logic.
4. Helps in **responsive design**.
5. Example:

```
if (/Android|iPhone/i.test(navigator.userAgent)) {  
    alert("Mobile Device");  
}
```

Q4 What is the purpose of clone() in DataSet methods?

Answer:

1. **clone()** copies the structure of the DataSet.
2. It does **not copy actual data**.
3. It copies tables, columns, and schema.
4. Useful when you need a **same structure empty dataset**.
5. Helps in creating backup structure.

Q5 What is JSON?

Answer:

1. JSON stands for **JavaScript Object Notation**.
2. It is used to **store and exchange data**.
3. Data is stored in **key-value pairs**.
4. It is lightweight and easy to read.
5. Commonly used in **web APIs and applications**.
- 6.

Q1.6 What is SOAP?

Answer:

1. SOAP stands for **Simple Object Access Protocol**.
2. It is used for **communication between applications**.
3. Works on **XML format**.
4. It is a **protocol for web services**.
5. It is more secure but heavier than JSON.

Q2. Answer the following questions. (2x15=30)

Q1 What is static method of JavaScript? Write program to calculate area and circumference of a circle

Answer:

❏ Static Method in JavaScript

1. A static method belongs to a **class, not objects**.
2. It is called using the **class name**.
3. It is used for **utility functions**.
4. It does not need object creation.
5. Example: `ClassName.methodName()`

❏ Program: Area & Circumference of Circle

```
class Circle {  
  
    // Static method for area  
    static area(radius) {  
        return Math.PI * radius * radius;  
    }  
  
    // Static method for circumference  
    static circumference(radius) {  
        return 2 * Math.PI * radius;  
    }  
}  
  
// Example  
let r = 5;  
  
console.log("Area: " + Circle.area(r));  
console.log("Circumference: " + Circle.circumference(r));
```

Q2 What is C# Exception Handling? Write program for Armstrong Number

Answer:

☐ C# Exception Handling

1. Exception handling is used to **handle errors** in program.
2. Prevents program from crashing.
3. Uses **try, catch, finally** blocks.
4. Helps in **safe execution** of code.
5. Improves program reliability.

Structure

```
try {  
    // risky code  
}  
catch {  
    // handle error  
}  
finally {  
    // always runs  
}
```

Program: Armstrong Number

```
using System;  
  
class Program  
{  
    static void Main()  
    {  
        try  
        {  
            int num, sum = 0, remainder, temp;  
  
            Console.Write("Enter a number: ");  
            num = int.Parse(Console.ReadLine());  
  
            temp = num;  
  
            while (num > 0)  
            {  
                remainder = num % 10;  
                sum += remainder * remainder * remainder;  
                num = num / 10;  
            }  
        }  
    }  
}
```

```
        if (temp == sum)
            Console.WriteLine("Armstrong Number");
        else
            Console.WriteLine("Not Armstrong Number");
    }
    catch (Exception e)
    {
        Console.WriteLine("Error: " + e.Message);
    }
}
```

Explanation

1. Armstrong number = sum of cubes of digits.
2. Loop extracts each digit.
3. Cube of digits is added.
4. Compared with original number.
5. Exception handling avoids crash on invalid input.